Every single beta-eta reduction step is a parallel beta-eta reduction step

The Open Logic Project contributors

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Editorial note. All delivered mathematical text is the author's own, copied from the pinned Open Logic Project unit file sources/olp-lambda-calculus-church-rosser/pinned-original-source.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: the eta-contraction case with its reflexivity citation and the explicit beta-case delegation are both verbatim in the span.

Editorial note. The label lem:one-par names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own lem environment in the unit source.

Editorial note. OLP reuses this label inside the unit; the card is identified by its statement span (source lines 433-435) and the claim key carries the statement start line.

Allowed standard prerequisites (editorial).

- Defining rules of one-step and parallel beta-eta reduction. Same unit, \ollabel{defn:beredone} (from the syntax unit's eta-reduction definitions) and \ollabel{defn:beredpar}. The eta-contraction case unfolds the parallel definition's eta rule; reflexivity supplies the operands.
- Reflexivity of parallel beta-eta reduction. Same unit, \ollabel{thm:refl} of the parallel-beta-eta section (source line 776). Every subterm may be left unreduced in the parallel step, giving the identity sides of the eta contraction.
- The beta-section bridge lemma. Same unit, \ollabel{lem:one-par} of the beta-reduction section (source line 486). The beta cases of the derivation induction are delegated with 'The other cases are as in' that lemma.

Source reference. Original theorem thm-refl: sources/olp-lambda-calculus-church-rosser/pinned-original-source.tex, source lines 790–790. Prerequisite pointer invoked at proof line(s) 440; it names a standard fact of the same unit that the

counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original definition defn-beredone: sources/olp-lambda-calculus-church-rosser/pinned-original-source.tex, source lines 439–439. Named citation invoked at source line(s) 439: the author's marker \olref[syn][eta]{defn:beredone} resolves inside the full Open Logic Project to another unit (not staged in this corpus); the cited fact is used as a standard result and every step of the counted proof built on it is retained verbatim in the delivered body.

Lemma 0.1. *If $M =_\beta M'$, then $M \to_\beta M'$.*

Proof. By induction on the !!derivation of $M =_\beta M'$. If $M =_\beta M'$ by η-conversion (i.e., defn-beredone), we use thm-refl. The other cases are as in 0.1. □